

Unified Master Architecture of the Resonant Informational Manifold (RIM): A Complete Fundamental Theory in Formal Mathematical Form

Abstract

This document provides the fully unified mathematical formulation of the Resonant Informational Manifold (RIM), integrating the darkon-mirror field cosmology, emergent spacetime dynamics, quantum-classical correspondence, elastic/fractal time, and matter-field resonance unification. It serves as the complete master ontology and governing equation set.

1 Foundational Postulate

All physical phenomena emerge from a single complex-valued resonant-information field:

$$\Phi(x^\mu, \tau) \in \mathbb{C}, \quad (1)$$

where x^μ is 4D spacetime and τ is fractal-elastic meta-time.

2 Master Field Equation

$$\mathcal{D}_\tau \Phi = Q \left[\square_g \Phi - \lambda |\Phi|^2 \Phi + \alpha \mathcal{F}(\omega, t) + \beta \rho_{DM}(r) + \gamma T_{\mu\nu} T^{\mu\nu} \right], \quad (2)$$

with constants $(Q, \lambda, \alpha, \beta, \gamma)$ fixed by resonance invariants.

3 Spacetime Emergence

$$g_{\mu\nu} = f_1(|\Phi|) \eta_{\mu\nu} + f_2(\partial_\mu \Phi \partial_\nu \Phi^*). \quad (3)$$

4 Darkon Density and Galaxy Rotation

Darkon density is expressed as

$$\rho_d(r) = d \rho_0 \left(\frac{r}{r_s} \right)^{-\gamma} \left[1 + \left(\frac{r}{r_s} \right)^\alpha \right]^{(\gamma-\beta)/\alpha}, \quad (4)$$

where d is the darkon unit.

Enclosed mass:

$$M(< r) = 4\pi \int_0^r r'^2 \rho_d(r') dr'. \quad (5)$$

Circular velocity:

$$v_c(r) = \sqrt{\frac{GM(< r)}{r}}. \quad (6)$$

5 Quantum-Classical Bridge

The wavefunction is not ontic; instead

$$\psi = \Pi[\Phi], \quad (7)$$

with Π a projection operator reducing informational phase-space dimensionality.

6 Elastic-Fractal Time

Meta-time obeys

$$d\tau = a(t) dt + b(t) dt^{\alpha(t)}, \quad (8)$$

with $0 < \alpha < 1$.

7 Master Action

$$S = \int d^4x \sqrt{-g} \left[R + |\nabla\Phi|^2 - \lambda|\Phi|^4 + U_{DM} + U_{mirror} \right]. \quad (9)$$